Fanjia Yan

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Education

University of California, Berkeley Class of 2025
M.S. in Electrical Engineering & Computer Science
University of California, Berkeley Class of 2024
B.A. in Computer Science and Applied Mathematics
Relevant Coursework:
Natural Language Processing, Deep Learning, Operating Systems, Machine Learning, Algorithms, Data Structures, Optimization Models, Computer Architecture

Experience

Member of Technical Staff, Microsoft AI **Nov 2026 – Now**
Applied Scientist, Amazon AGI **May 2025 – Sep 2026**

- Nova 2 Lite - **Core Contributor**
  - Led Nova 2 Lite **Agentic RL** stage production trainings on 8K H200s, including recipe mixing ablations, and supporting RL rollout infra with 10+ external environments and thousands of concurrent asynchronous rollouts
  - Designed agent SFT/RL **training data recipes** with tool use spanning general tool calling, stateful user-agent interaction, web search, vending and office agent tasks
  - Enhanced in-context policy **memory management** during RL rollout by adding a structured reasoning prefix that encouraged retrieval of earlier strategies, improving performance on long-horizon, stateful decision tasks.
  - Drove adoption of the first pre-training agent evaluation suite, and Doubled the internal post-training **agent evaluations** suites with rapidly adopting open source benchmarks, 3rd party labeling data, and Amazon subsidiaries(AWS, Alexa) customers' use cases, improving generalization and evaluation coverage on unseen tasks

Graduate Researcher, Berkeley Sky Computing Lab **Aug 2023 – May 2025**

- Berkeley Function calling Leaderboard**(BFCL) – Project Lead**
  - Directed a year-long effort to unify industry evaluation standard for tool calling and curate 5K high-quality dataset that captures structural agent behaviors resulting in the most widely adopted tool calling benchmarks to date
  - Trained **Openfunctions**, a 6.91B parameter function calling model fine-tuned with 65K synthetic **tool calling** QA pairs from real world usage, which received 12K deployments within a month on HuggingFace.
- LiveCodeBench **– Core Contributor**
  - Designed automatic evaluation platform that periodically collects coding problems from latest programming contests.
  - Built **ExecEval**-based evaluation system that handles concurrent code executions at scale.

Intern of Technical Staff, Cohere **Jan 2025 – May 2025**

- Command A - **Contributor**
  - Implemented and trained Medusa **Speculative Decoding** head on Command A for Cohere inference API, accelerated code generation by 3x and creative writing by 2x
  - Enabled **sequence packing** and improved Speculative Decoding head training mixture, leading to 4x training speed up while maintaining same decoding efficiency

Deep Learning Software Engineer Intern, Nvidia **May 2024 – Aug 2024**

- Prototyped Nvidia Drive in-car assistant by building a multi-modal **Retrieval-Augmented Generation** (RAG) pipeline for retrieving car manual knowledge using **NV-Embed** and **NeVA-22B**, which was presented at monthly demo
- Fine-tuned **Llama-3.1** with **LoRA** on efficient car control tool usages and SERP web browsing, significantly boosting in-car assistant tool calling accuracy and web search capabilities

Selected Publication (Google Scholar)

1. Shishir G. Patil, Huanzhi Mao, **Fanjia Yan**, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, Joseph E. Gonzalez. The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models. (Accepted to ICML 2025)
2. Naman Jain, King Han, Alex Gu, Wen-Ding Li, **Fanjia Yan**, Tianjun Zhang, Sida Wang, Armando Solar-Lezama Koushik Sen, Ion Stoica. LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code. (Accepted to ICLR 2025)

Open Source Project

1. Gorilla (Contributor) 12.6K Stars
   An open source initiative to build full stack LLM training and evaluation software for Tool Usage
2. Aqueduct (Contributor) 521 Stars
   An open source MLOps framework that run LLM and ML workloads on any cloud infrastructure